Quality Control Protocol - Leica Confocal Microscopes

Contents

Necessary equipment: 1
Start-up: 2
Cleaning: 3
Check Z flatness of stage: 3
Laser Power: 4
Detector Dark Noise: 5
Detector Gain Linearity: 6
Field illumination homogeneity: 6
Fluorescent bead imaging for PSF: 7
Fluorescent bead imaging for multichannel PSF: 8
Check X-Y calibration of stage via stitching accuracy: 9
Check secondary objectives: 10
Shut-down: 10
Analysis 10
Excel 10
ImageJ 11
MATLAB 11
Python 12
Finishing up 12

Necessary equipment:

- Printed copy of laser power template for microscope
  - Or: laptop to fill in the Excel template
- If needed: copy of Koehler Illumination protocol
- Thorlabs laser power meter PM100D
- Thorlabs power meter slide-shaped sensor S170C
- Z-stage (to swap out with Z-galvo, if Z-galvo is normally installed)
  - Not possible at: SP8, Upright Stellaris
- Sample to image with transmitted light path (i.e., Golgi-stained brain slice)
- Chroma slides
- Stained cells slide (Abberior or GattaQuant)

- Ceramic dots slide
- Pipettes
- Set of hex keys
- Fluorescent beads: 170nm green (505nm emission), 500nm multicolor (430+515+580+680 emission)
    - Check that beads are not older than one year
    - For protocol to make beads, see: J:\Methods and Protocols\Sample preparation\ FluorescentBeads_updated250324.docx
- For cleaning:
    - Zeiss lens cleaning solution (Isopropanol + n-Hexane)
        - For recipe, see: J:\Methods and Protocols\Cleaning and maintaining optics\ The_Clean_Microscope_Zeiss.pdf
    - Dust blower
    - Cotton-tipped wooden sticks
    - Distilled water
    - Front Lens Imager toolbox

<u>Start-up:</u>

- Create a sub-folder with current date in:

  J:\Equipment\Microscopes\Imaging Facility**[microscope]**\QualityControl sub-folder name: yymmdd

- Switch system on, ensure that laser key is turned to correct position
- Start LAS-X software
- In Configuration → Hardware, set bit depth to 16 bit
- In Configuration → Laser Config, activate all lasers
- Set laser power of WLL to 85 %, Argon to 60%
- Let the instrument warm up for 60 min before start of QC measurements
- If the microscope is fitted with Z-galvo:
    - Remove it, and install the normal z-stage
    - Check the flatness of the sample holder associated with the Z-galvo by eye
    - **NOTE**: This is not currently possible at the SP8, due to the relative positions of stage and objectives; can't focus on a slide mounted in the normal z-stage with the 10x.
    - **NOTE**: This is also not possible at Stellaris upright; the "normal" z-stage is too small

- Mount the transmitted light sample and align condenser for Koehler illumination according to protocol: J:\Methods and Protocols\Quality control\KoehlerIllumination_SetUp.docx
    - **NOTE**: Not possible at SP8 or upright Stellaris because of relative positions of condenser, stage, and objective turret.
- Note temperature in the room if possible:
- Check drives on the PC: Are they almost full?
    - Delete data older than 1 year
    - Make a list of file owners with substantial amounts of data, to ask if it can be deleted

---

Cleaning:

- Clean all objectives, the condenser lens and oculars:
    - Use the dust blower to remove loose particles from front and back lenses
    - Use distilled water and a cotton swab to remove water-soluble dirt
    - Use Zeiss Brillenreiniger and a cotton swab to remove oily dirt by moving swab in an outwards spiral
    - Inspect lens; if it is not yet clean, use Zeiss cleaning solution and a cotton swab. Again, use an outward spiral motion, from the center to the edge of the lens.
- Check all objectives for signs of damage:
    - Use inverted ocular to inspect the front lens and back lens
    - Use Front Lens Imager to make an image of each objective. Label it, and place it in a folder that will be move to today's QC folder at the end.
    - If not possible to use Front Lens Imager, use phone camera to take images of the front lens of each objective
- Wipe dust and any spills from microscope and surrounding area
- Clean touch screen
- Check that there's enough (and refill if needed):
    - immersion medium
    - EtOH
    - glasses cleaner
    - lens wipes
    - Kim wipes

Check Z flatness of stage:

Note: This segment may be completed before the lasers have fully warmed up; i.e., within the first hour of turning on the system.

Sample: ceramic dot slide

Objective: 10x air, low NA

Procedure:

- Find the focal plane using epifluorescence.
- Mark four dots in LAS-X navigator using settings for Alexa488 or similar:
    - one in each corner of the slide or field of view
    - name them topleft, topright, bottomleft, bottomright
- Set a z-stack that encompasses the majority of all four dots; use a 4µm Z-step
- Image a stack of each dot, using the setting “Same stack for all regions”
- For Z-stage without Z-galvo:
    - If the stacks are not centered, adjust stage using hex key to lower or raise one side
    - Repeat imaging and adjusting until stacks align with each other
- With Z-galvo:
    - Locate the focal plane in
    - Note the shift in micrometers:
        - Left to right:
        - Top to bottom:
- Saving data is not necessary for this segment. If you save it, label the LIF: stage_flatness

---

<u>Laser Power:</u>

Sample: ThorLabs power meter slide S170C

Objective: 10x air, low NA

Pixels: 64 x 64

Speed: 10

Zoom: Highest

Bidirectional scanning

Detector "on"

Procedure:

- Mount Thorlabs Laser Power Meter sensor S170C
    - Do not focus the laser on the sensor, because focus does not influence power measurement, but may damage the sensor
- Use highest possible zoom
- Switch off room light
- For each laser, wavelength, and percentage on the template sheet:
    - Set the power meter to the correct wavelength
    - On the power meter, press Δ twice to set the reference value
    - Adjust the laser's percentage in LAS-X
    - Make sure a detector is on, even though it isn't being used; otherwise, "Live" won't be enabled
    - Press "live" in LAS-X to start scanning
    - Record the power meter value in **µW** next to that wavelength/percentage on the sheet
- Toggle off Bidirectional Scanning in LAS-X
- Return to Zoom 1

<u>Detector Dark Noise:</u>

Sample: None

Objective: 63x

Pixels: 512 x 512

Zoom: 1

Speed: 400

Averaging: None

Procedure:

- Switch to 63x objective
- Close the pinhole as much as possible
- Open .lif file for detector dark noise from a previous QC to use as template

    - Acquire one multichannel image, with one channel per detector
- Check that all lasers are set to 0% in the “acquisition” tab
- Shut off room light
- Make a new project for today
- Make the acquisition
- Save project from today as “detector_darknoise.lif”
- Close both projects
- Put pinhole back to normal size; 1 AU

Detector Gain Linearity:

Sample: yellow-green Chroma slide

Objective: 10x air, low NA

Pixels: 512 x 512

Speed: 400

Averaging: 4 frames

Procedure:

- Open .lif file for detector gain from a previous QC to use as templates
- Make a new project for today
    - Acquire one image per detector, with 5 channels, with each channel being a different gain
        - For HyDs: 100, 200, 300, 400, 500
        - For PMTs: 200, 400, 600, 800, 1000
    - Adjust laser power so that image is bright, but not saturated, at the highest gain setting. The laser power should be the same for all detectors.
- Save project from today as “detector_gain_linearity.lif”
- Close both projects

Field illumination:

Sample: Yellow-green Chroma slide

Objective: ALL

Pixels: 1024 x 1024

Speed: 400

Zoom: 0.75

Averaging: 4 frames

Procedure:

- Open a previous field_illumination.lif for using settings
- For each objective, set a multi-channel image with the one channel per fiber input. Leica scopes have two fibers, UV and visible. Use 405 and 488.
    - Adjust laser power and detector gain to avoid saturation
    - Acquire image
- Save LIF as “field_illumination.lif” in J:\Equipment\Microscopes\Imaging Facility**[microscope]**\QualityControl\yymmdd\
- Close projects

<u>Fluorescent bead imaging for PSF:</u>

Sample: 170nm green fluorescent beads; 500nm multicolor beads

Objective: ALL, except 5x

Pixels: ~2048 x 2048; more may be needed to have enough beads in FOV with correct pixel size.

Speed: 400

Averaging: 4 lines

Procedure:

- For each objective with 20x magnification or higher:
    - Mount a slide with 170nm green fluorescent beads
    - Find FOV with beads separated and located mostly in the center.
        - Should have approximately 20 beads in FOV

    - Find out suggested XY pixel value, then triple it to oversample. Suggested sizes are:
        - 0.03µm for 63x
        - 0.045µm for 20x
    - Find out suggested z-step; halve the step size to double number of imaged planes. Suggested step sizes are:
        - 0.15µm for 63x
        - 0.5µm for 20x
    - Set a dark-to-dark z-stack. Estimated stack sizes to encompass the full PSF are:
        - 4µm for 63x
        - 12µm for 20x
    - Check that the image is not saturated in any channel
    - Image the stack
    - Name file: 63x_PSF_488
- For 10x objective:
    - Mount a slide with 500nm fluorescent beads
    - Repeat process for imaging a field of view with several beads
    - XY pixel: 0.080µm
    - Z-step: 1.2µm
    - Z-stack size: 40µm
- If a PSF appears asymmetric or otherwise indicative of a problem: redo the acquisition, with the pinhole completely open, to try to distinguish if the aberrations are due to objective or microscope.
- Save only one acquisition per lif
- Save each LIF in J:\Equipment\Microscopes\Imaging Facility\[microscope]\QualityControl\yymmdd\

<u>Fluorescent bead imaging for multichannel PSF:</u>

Sample: 500nm multicolor fluorescent beads

Objective: 63x

Pixels: 512x512

Speed: 400

Zoom: High, to ensure one bead per FOV

Averaging: 4 lines

Procedure:

- Open previous multicolor_beads image to apply settings
    - Line sequential, with one sequence for each of these wavelengths: 405, 488, 633
- Find FOV with a **single** bead
- Suggested pixel sizes:
    - XY: 0.03µm
    - Z: 0.15µm
    - Oversampling improves MetroloJ's processing of the data
- Set a dark-to-dark z-stack
- Check that the image is not saturated in any channel
- Image the stack
- Name file: 63xOil_multicolor_bead.lif

<u>Check X-Y calibration of stage via stitching accuracy:</u>

Sample: Stained cells (from Abberior or other)

Objective: 63x

Pixels: "Optimal" Nyquist sampling

Speed: 400

Bidirectional scanning

Zoom: 1

Averaging: 2 lines

Procedure:

- Mount a biological sample (i.e., cells stained with 4 dyes)
- If available, open "bio_sample.lif" and apply image settings
- Set up a 3 x 3 tile scan in 4 channels
- Set 3 to 5 Z-planes
- Turn ON "Keep raw data" in acquisition

- Acquire image
- Save as “bio_sample.lif”

Check secondary objectives:

Note: Microscope and PC must be completely shut down before removing the nosepiece/turret to change to a different nose piece. Individual objectives may be changed out while the microscope is on, by adjusting Hardware Configuration.

- Close the LAS-X software
- Open the software for Hardware Configuration
- Mount additional objectives that have not yet been tested, but are regularly used at this microscope (water dipping objectives, CLARITY objectives, etc.)
- Assign the objective to the corresponding port in Configure → Nosepiece by clicking on the port number, then typing the six-digit number into the search bar and double-clicking the entry matching your objective.
- For each objective, repeat the protocols for measuring the PSF with 170nm beads and for illumination homogeneity.

Shut-down:

- Record temperature in room:
- Move photos taken with Front Lens Imager into today’s QC folder
- Shut down the PC and microscope
- Switch back to Z-Galvo if it was removed
- Clean objectives
- Check stage and surrounding area for spills, dust, etc.

Analysis

💡 Do these steps in order, because some sections rely on previous sections (exports from ImageJ are needed for the Python code).

Excel

- Copy the laser power template into today's QC folder
- Fill it out with values recorded on the printed template

ImageJ

- Open the 'field_illumination' LIF
- Use File → Save as… → Tiff for each image; one per objective.
    - They should be labeled like the following: "field_illum_5x_405_488.tif"
    - Include the objective magnification, and each wavelength used, separated by underscores.
- Open the '63x_multicolor_bead' LIF
- Check that only one bead is visible in the image; crop it if necessary
- Open the 'metroloJ' Plug-in
    - We are using version 1.3.1.1.
- Choose the "Co-alignment" module (4th from the left, top row).
- Change the file name to coalignment_analysis.
- Set "Actual image bit depth" to 16
- In "Microscope Acquisition Parameters":
    - Set Microscope type to CLSM
    - Specify NA and RI of objective and immersion media
        - If unsure of the NA, check: J:\Equipment\Optics\Objectives\ObjectivesInventory.xlsx
    - List the excitation wavelengths for each channel
- The defaults for Bead Detection Options are generally fine
- Check "Apply tolerances to the final report"
- Set "Reject coregistration ratio above" to 1.0.
- Click OK.
- In today's QC folder, a "Processed" folder will be created by metroloJ. Move the PDF out of this folder, into today's QC folder, and delete the "Processed" folder.
- Open "bio_sample.lif"
    - Inspect for quality. Check that tiles are well-aligned by adjusting brightness/contrast such that black rectangles at the edges are visible. These indicate that the stage did not move exactly parallel to scanning.

MATLAB

- Make sure that the folder J:\Equipment\Software\MATLAB and all subfolders are listed in "Home → Set path"
- In the "current folder" window, navigate to today's QC
- For each single-channel PSF LIF:

- Run the following line of code (values are: filename, showPlot, zoomIn, redThreshold, showMicroscopeEvalPlot, showZProjection, beadSize, extractMetaDataFromPath)

psfevalleica('63xOil_PSF_488.lif',1, 1, 1.7, 1, 1, 0, 1);

- Save the figure showing “theo, ortho, pca” and the three mean bead images as a PNG titled “63x_PSF.png”

Python

- Open Python script “QC_functions.py” and run it so that the functions are accessible by the Leica QC script.
- Open Python script “QC_plots_Leica.py”
  - **Modules**: laser power, detector gain linearity, detector dark noise, field illumination, stage x-y repositioning
- Change line 21 to indicate path to current QC folder, for the correct microscope and date:

  # Input the path to QC folder for a particular date:
  qc_path = r"J:\\Equipment\\Microscopes\\Imaging Facility\\Leica_SP8_N131\\QualityControl\\250321”

- Run the script.
- Check the QC folder for the following files:
  - laser_power.png
  - detector_gain_linearity.png
  - detector_darknoise.png
  - field_illum…..png
  - PNG for each objective for the PSFs

Finishing up

- Copy/paste the Quality Control Results Summary Template to today’s QC folder.
- Insert all PNGs into the correct place of the template.
- Make notes of any follow-ups that should be done at the top of the template.
- If applicable, make a folder in the “LogFiles” folder of the microscope for issues that need to be followed up on.
  - Fill out a “Template - Hardware issues” Word Doc for each issue.
  - Email the microscope company if needed.

- Record today's date in this Excel sheet: J:\Equipment\Microscopes\Imaging Facility\ QC_Schedule.xlsx to track how often quality control is being performed.
- Email the users who have too much data stored locally on the system and ask if they can delete the data as soon as possible, or if the facility can delete the data for them.